

Homework 9

40 points

1. From 10 men and 8 women, a committee of 6 is chosen. How many possible committees contain **at least 3 women**? **5 points**

2. From 12 distinct books you want to choose an ordered reading list of 4 books (order matters). How many ordered lists have **a specific book** *B* included **somewhere** in the list? **5 points**

3. How many **4-digit numbers** can be formed using digits 1–6 if **repetition is allowed**, but the number must be **even**? **5 points**

4. The word “PROBABILITY” has 11 letters (vowels: O, A, I, I).

If the letters are arranged at random, find the probability that **all vowels are together**. **5 points**

5. How many **3-digit numbers** formed using digits 1–9 (no repetition) are **divisible by 5**?

What's the probability that a random 3-digit number from these digits is divisible by 5? **$5 + 5 = 10$ points**

6. A player wins each match with probability $p = 0.6$. He plays until 5 wins. What is the probability that he **finishes (gets 5 wins) on or before the 9th match?** **5 points**

7. A basketball player has a 70% free throw success rate.

If she takes 5 shots, find the probability she makes **at least 4 baskets**. **5 points**